For international companies: available via legal entities in Cyprus, Georgia, or Dubai. Deel also works.

Arkadiy Pechnikov

Senior Python Backend Engineer | Crypto & Trading Infrastructure

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PROFESSIONAL SUMMARY

Python backend engineer who builds systems that actually handle load. Scaled a Temporal.io platform from ~10 to 200 jobs/sec (20x), reverse-engineered a WASM captcha in Ghidra and rewrote it in Rust for a 20x speedup, and built an SDK generation pipeline that cut exchange integration from ~1 month to ~1 week across 20+ undocumented CEX APIs. Comfortable with Kubernetes, full observability stacks (Grafana/VictoriaMetrics/Loki), and enough reverse engineering - WebSocket protocols, smart contracts, WASM binaries - that picking up new tech is usually the easy part. After running a 9-person team as founding CTO, I'm focused on deep IC work inside a strong engineering org.

TECHNICAL SKILLS

Languages: Python, Rust, JavaScript/TypeScript, SQL
Frameworks & Libraries: FastAPI, aiohttp, asyncio, Celery, Temporal.io, Pydantic, SQLAlchemy
Databases: PostgreSQL, Redis, MySQL, Elasticsearch, TimescaleDB
Cloud & DevOps: Kubernetes, Docker, FluxCD, GitHub Actions, CI/CD, Helm, Cloudflare, MinIO, NixOS, k3s, Terraform
Observability: Grafana, Victoria Metrics, Loki, Sentry
Blockchain & DeFi: Web3.py, ethers.js, EVM, DeFi protocols (Uniswap, 1Inch, Stargate, LayerZero), TON SDK, mitmproxy
Security & Analysis: Web Application Analysis (Burp Suite, Chrome DevTools), JavaScript Deobfuscation, WebSocket Protocol Analysis, On-chain Transaction Analysis, ABI Encoding/Decoding, Traffic Analysis (mitmproxy), WASM Binary Analysis (Ghidra)

PROFESSIONAL EXPERIENCE

Senior Backend Engineer (Contract)

Fenixwb - Wildberries slot booking service - 200 jobs/sec, 10K+ users | Oct 2024 - Sep 2025 | Remote

- Inherited an unstable monolith prototype pushing ~10 tasks/sec - redesigned the entire backend on Temporal.io with ~10 microservices integrated with Elasticsearch and MySQL
- Pushed throughput from 10 to 200 background jobs/sec (20x), the threshold that let the product handle real user load and start onboarding paying customers
- Disassembled a WASM captcha binary in Ghidra, then rewrote the solver in Rust - execution dropped from 2s to 0.1s (20x faster), giving clients higher booking success rates than competing bots
- Migrated from MongoDB to PostgreSQL - simpler queries, fewer ops headaches for a 3-person backend team
- Traced a critical throughput bottleneck via hypothesis-driven load testing - the root cause fix is what took us from ~10 to 200 tasks/sec
- Went from zero monitoring to production-grade observability: Grafana + Victoria Metrics + Loki with custom dashboards and alerting - incidents detected in ~3 minutes vs previously hours, MTTR dropped from days to hours

Tech: Rust, Python, Temporal.io, Elasticsearch, MySQL, PostgreSQL, asyncio, Grafana, Victoria Metrics, Loki, Docker, Kubernetes

Senior Python Developer

Cryptosofters - Multi-wallet orchestration platform with visual pipeline builder - near-production, drag-and-drop wallet operations | May 2023 - Nov 2024 | Remote (Dubai-based)

- Designed multi-wallet orchestration platform with drag-and-drop pipeline constructor - users configured chains of on-chain operations (swaps, bridges, contract calls) without code. Built to near-production before investment stopped
- Under the hood: event-driven state machine with per-wallet operation queues, automatic retry with exponential backoff, and transaction rollback on partial failures - the engine that made drag-and-drop orchestration reliable enough for real money
- Zero security incidents across the platform's lifetime. Architected per-user wallet isolation, credential management, and transaction signing in a multi-tenant environment where investor confidence hinged on it
- Reverse-engineered closed-source smart contracts via on-chain transaction analysis and ABI decoding - often the only way to integrate protocols with no SDK, no docs, and no public API
- Architected the SDK generation pipeline: mitmproxy traffic capture → OpenAPI spec → auto-generated Python SDK for 20+ CEX APIs. Cut exchange integration from ~1 month to ~1 week (75%) - the force multiplier that made a 9-person team viable on a $100K budget
- Delivered 20+ DeFi protocol integrations (Uniswap, 1Inch, Stargate, LayerZero) with a unified interface via web3.py - per-protocol time dropped from ~1 week to ~1 day (~80%), and any team member could add protocols, not just the specialists

• Introduced code review process and CI/CD pipelines for a team of 9 - regression bugs dropped noticeably and new engineers onboarded faster

Tech: Python, Web3.py, mitmproxy, OpenAPI, asyncio, PostgreSQL, Redis, Docker

Middle+ Python Developer / Tech Lead

Vitrinagram - Telegram MiniApp e-commerce platform - multi-tenant storefronts, 1,000+ concurrent users per store | Nov 2021 - Apr 2023 | Remote (Moscow-based)

• Built multi-tenant Telegram MiniApp backends handling 1,000+ concurrent users per store - horizontal scaling via Celery + Kubernetes meant cost grew linearly with tenants, not exponentially

• Automated tenant provisioning: FluxCD + Cloudflare API pipeline, ~3 minutes from commit to live subdomain

• Developed an S3-compatible proxy API on MinIO for tenant-isolated file storage - eliminated the shared-storage security risk and cut costs vs managed S3

• Per-tenant observability out of the box: Grafana + Victoria Metrics + Loki + Sentry dashboards clients could use for self-service debugging - ~40% fewer support tickets once clients could self-debug

Tech: Python, FastAPI, Celery, PostgreSQL, Redis, Docker, Kubernetes, FluxCD, Grafana, Victoria Metrics, Loki, Sentry, MinIO, Cloudflare

Python Backend Developer

Freelance - Crypto infrastructure, trading systems, and data pipelines | Jan 2018 - Present

• Built EVM wallet management system for a client with multiple investors: 2,000+ wallets, 5,000+ tx/day, fully autonomous - equivalent of eliminating a full-time ops person

• Reverse-engineered TradingView's WebSocket protocol and deobfuscated their JavaScript - collected 260M candlesticks in ~1 week covering 20+ years of forex and crypto data. Comparable feeds charge $600+/month

• Led 3 Go developers building a CEX arbitrage bot with sub-100ms execution latency - cross-language coordination (Python orchestration, Go hot path) where every millisecond was the difference between profit and a missed window

• Portfolio managers needed consistent exposure across 10+ exchange accounts - built an automated replication system mirroring positions in real-time with near-zero drift between source and mirror

• Developed a USDT-TON payment gateway with near-instant confirmation for a gaming platform - settlements in seconds vs traditional processing delays (delivered, pre-launch)

• Chose Rust for the Polymarket transaction parser because Python couldn't sustain the throughput - 300 blk/s from distributed event sources, production data ingestion

• Deployed single-node k3s on NixOS with declarative config, Terraform provisioning, and GitOps workflow - reproducible infra, entire cluster from zero in under an hour

• Provisioned Kubernetes clusters on Yandex Cloud and Hetzner using Terraform for multiple clients - standardized the infra setup, cut provisioning time across engagements

• Building quantitative backtesting infrastructure on nautilus-trader with 260M+ candles stored in TimescaleDB - time-series queries that would choke PostgreSQL run in seconds

Tech: Python, Rust, JS/TS, Web3.py, ethers.js, PostgreSQL, Redis, asyncio, NixOS, k3s, Terraform, TimescaleDB, Docker

PROFESSIONAL DEVELOPMENT

Machine Learning Specialization - Stanford University (via Coursera) (2026)

Reinforcement Learning for Trading Strategies - New York Institute of Finance (2026)

Using Machine Learning in Trading and Finance - New York Institute of Finance (2026)

LANGUAGES

Russian: Native | English: Upper-Intermediate (B2)